1. Bedrooms: Holding other variables constant, 1 more bedroom decreases price by about $45,190.

2. Bathrooms: Holding other variables constant, 1 more bathroom increases price by about $42,650.

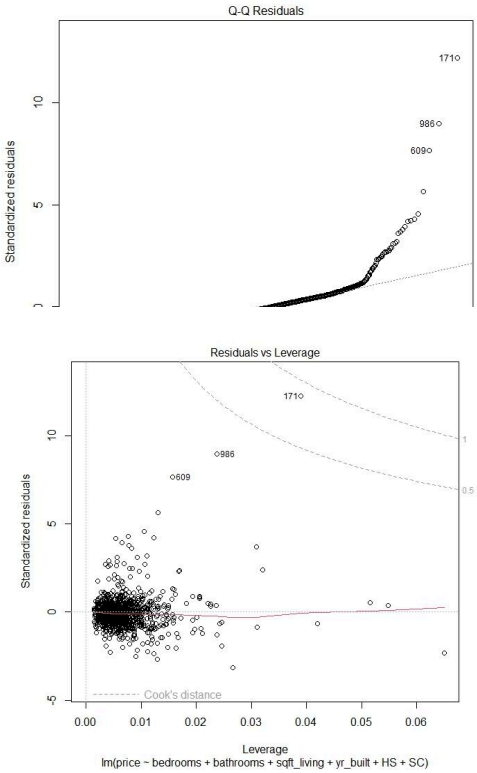
3. Sqft_living: Holding other variables constant, 1 more sqft increases price by about $268.

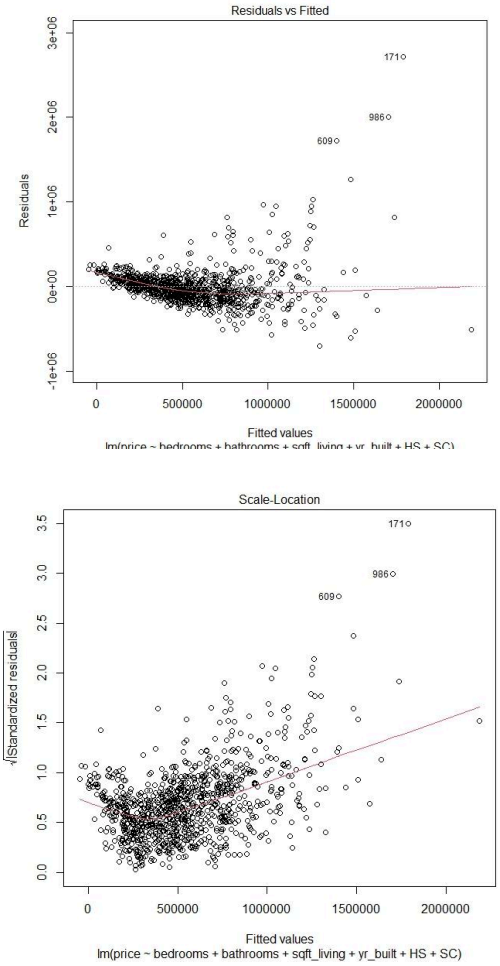
4. Yr_built: Holding other variables constant, 1 year newer decreases price by about $1,857.

5. HS: Holding other variables constant, an increase in HS decreases price by about $730,400.

6. SC: Holding other variables constant, an increase in SC decreases price by about $1,139,000.

The diagnostic plots show several problems with the model. There is heteroskedasticity since the spread of residuals increases, especially in the Scale-Location plot. There is also non-linearity, as the residual trends are not flat. The Q-Q plot shows that residuals are not normally distributed. Additionally, there are outliers and influential points visible in the plots. These issues suggest the model may not be the best fit.





To improve the base model, I transformed the dependent variable by using log(price) and also applied a log transformation to sqft_living. These changes were made to reduce heteroscedasticity and improve linearity, which were issues in the original model. I considered different model forms, but this final model provided the best balance between model fit and meeting the regression assumptions.
The improved model has an R^2 of 0.7709 and an adjusted R^2 of 0.7697, meaning it explains about 77% of the variation in housing prices. The model is statistically significant overall ($p < 0.001$), showing strong explanatory power.
There are no multicollinearity issues, as all VIF values are below 3.
The diagnostic plots show improvement compared to the base model. The Residuals vs Fitted plot shows a more random pattern, indicating better linearity. The Q-Q plot is closer to a straight line, suggesting residuals are approximately normal with slight deviations at the tails. The Scale-Location plot shows reduced spread, although there is still mild heteroscedasticity. The Residuals vs Leverage plot shows a few higher leverage points, but none are highly influential.

1. Bathrooms: Holding other variables constant, 1 more bathroom increases price by about 6.86%.

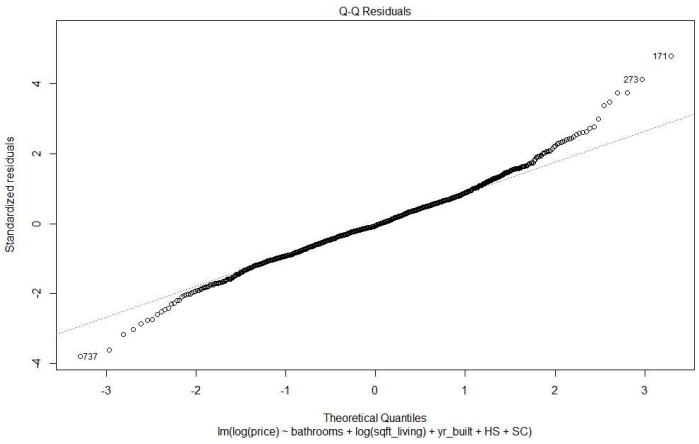
2. Sqft_living: Holding other variables constant, a 1% increase in sqft_living increases price by about 0.6278%.

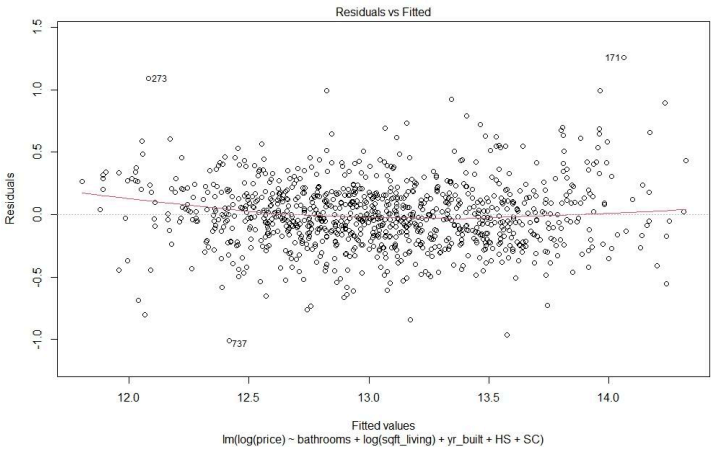
3. Yr_built: Holding other variables constant, 1 year newer decreases price by about 0.197%.

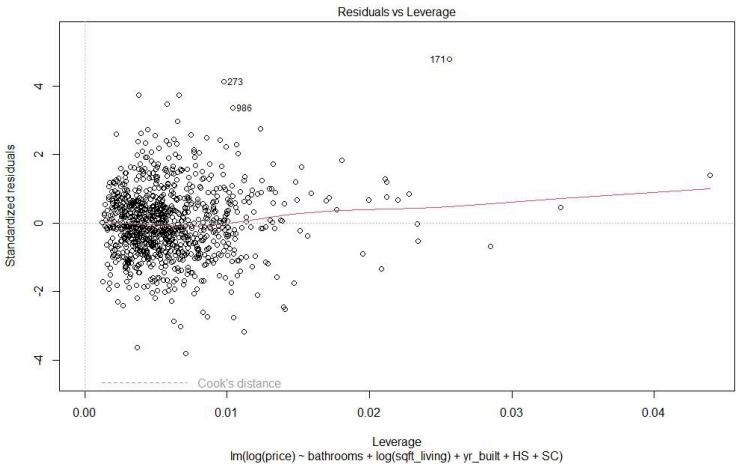
4. HS: Holding other variables constant, homes with HS = 1 have prices about 89% lower than homes with HS = 0.

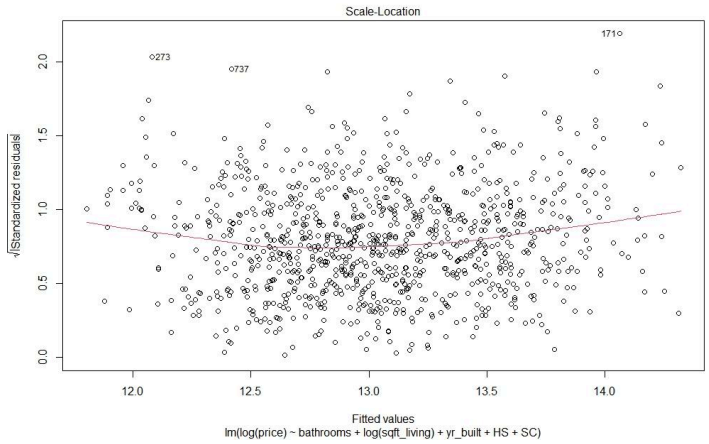
5. SC: Holding other variables constant, homes with SC = 1 have prices about 81% lower than homes with SC = 0.

Compared to the base model, this improved model better satisfies the regression assumptions while maintaining strong explanatory power. Overall, the final model provides a more reliable fit for explaining housing prices.


Q-Q Residuals
Standardized residuals
Theoretical Quantiles
lm(log(price) ~ bathrooms + log(sqft_living) + yr_built + HS + SC)
171
273
737


Residuals vs Fitted
Residuals
Fitted values
lm(log(price) ~ bathrooms + log(sqft_living) + yr_built + HS + SC)
171
273
737


Residuals vs Leverage
Standardized residuals
Leverage
lm(log(price) ~ bathrooms + log(sqft_living) + yr_built + HS + SC)
Cook's distance
171
273
986


Scale-Location
√|Standardized residuals|
Fitted values
lm(log(price) ~ bathrooms + log(sqft_living) + yr_built + HS + SC)
171
273
737